jupyter 8-Puzzle Astar Last Checkpoint: 1 hour ago

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Code

JupyterLab Python 3 (ipykernel)

```
display_path(path2)
```

```
A* using Misplaced Tiles
Number of moves: 2

Step 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 2
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

A* using Manhattan Distance
Number of moves: 2

Step 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 2
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

[3]:

```
def heuristic(state, goal):
    count = 0
    for i in range(9):
        if state[i] != 0 and state[i] != goal[i]:
            count += 1
    return count
```

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localhost:8888/notebooks/8-Puzzle%20Astar.ipynb


```
if solution:
    print("Solution Found!")
    print("Number of moves:", len(solution) - 1)

    for i, state in enumerate(solution):
        print("Step", i)
        display(state)
else:
    print("No solution found.")
```

```
Initial State:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Goal State:
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

Solution Found!
Number of moves: 2
Step 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 2
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

[4]:

```
# 8-Puzzle using A* Algorithm
# g(n) = Depth of node
# h(n) = Manhattan Distance
# f(n) = g(n) + h(n)
```

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jupyter 8-Puzzle Astar Last Checkpoint: 1 hour ago

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Code

JupyterLab Python 3 (ipykernel)

```
solution = astar(start, goal)

if solution:
    print("Solution Found!")
    print("Number of moves:", len(solution) - 1)

    for i, state in enumerate(solution):
        print("Step", i)
        display(state)
else:
    print("No solution found.")
```

```
Initial State:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Goal State:
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

Solution Found!
Number of moves: 2
Step 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 2
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

[ ]:

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